Healthcare Resource Availability: Analyzing Global Disparities

ADAN8880 - Directed Practicum in Analytics

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Date: March 2, 2025

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Week 6 Written Report1

Model approach:

For this week's prediction task, we selected a Random Forest Regressor (RF) as our modeling approach. The choice was motivated by several factors:

- **Robustness to Non-Linearity:** Life expectancy is influenced by multiple non-linear factors, and RF can handle complex interactions well.
- **Feature Importance Analysis:** RF provides insights into feature importance, which aids in understanding key determinants.
- **Handling of Missing Data:** RF can handle missing values and imbalanced data distributions better than traditional regression models.
- **Generalization:** By tuning hyperparameters, RF can be optimized to generalize well without overfitting.

Performance Metrics

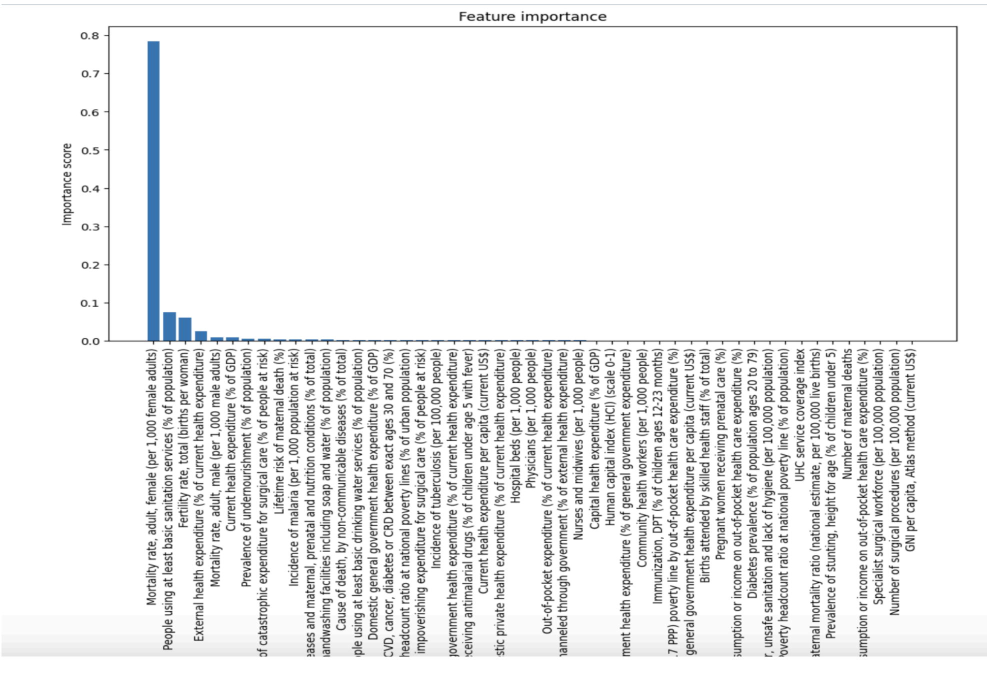
To evaluate model performance, we used RMSE and R^2 as our primary metrics.

- RMSE measures the average prediction error in the same unit as life expectancy, which makes it an intuitive way to assess how far predicted values deviate from actual values.
- R^2 quantifies how well the independent variables explain variations in life expectancy. A higher R^2 suggests the model captures most of the variance in the target variable, but an excessively high R^2 may indicate overfitting.
- We choose these metrics because they provide a balance between interpretability and effectiveness. RMSE gives a direct measure of prediction error, R^2 helps assess whether the model is too closely fitted to the training data. Alternative metrics like MAE could have been used, but RMSE was preferred because it penalizes larger errors more heavily, which is important in a health-related prediction task where significant deviations in life expectancy predictions can have serious implications.

Complexity of the Modeling Approach

- RF is a bagging-based ensemble learning technique that builds multiple decision trees and aggregates their predictions. The complexity is driven by:
- Number of Trees (n_estimators): More trees improve stability but increase computational cost.
- Tree Depth (max_depth): Deep trees capture fine-grained relationships but risk overfitting.
- Minimum Samples per Split/Leaf: Higher values reduce model complexity and prevent small, unreliable splits.

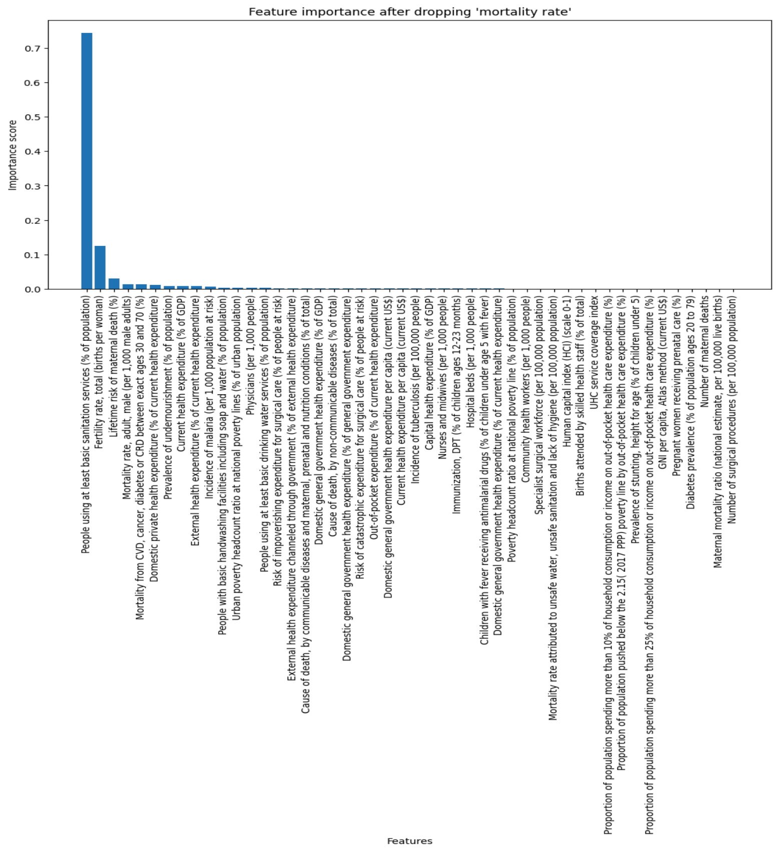
To balance predictive power and generalization, we tested three variations of the model.




```
Mortality rate, adult, female (per 1,000 female adults)                    0.78504
5
People using at least basic sanitation services (% of population)          0.07381
0
Fertility rate, total (births per woman)                                   0.05986
9
External health expenditure (% of current health expenditure)              0.02371
8
Mortality rate, adult, male (per 1,000 male adults)                        0.00861
0
Current health expenditure (% of GDP)                                      0.00831
8
Prevalence of undernourishment (% of population)                           0.00508
5
Risk of catastrophic expenditure for surgical care (% of people at risk)   0.00463
0
Lifetime risk of maternal death (%)                                        0.00384
8
Incidence of malaria (per 1,000 population at risk)                        0.00376
3
dtype: float64
```

Before making any adjustments, we first analyze feature importance to identify potential data leakage or any dominant variables. Based on the plot, "mortality rate, adult female" had an extremely high correlation with life expectancy, which may indicate feature leakage.

The feature "mortality rate, adult female" exhibited a strong inverse correlation (-0.9597) with the target variable, making it a nearly direct predictor of life expectancy. This high correlation posed a risk of target leakage, where the model could overly rely on this single feature rather than learning meaningful patterns. To ensure a more balanced and generalizable model, the feature was removed from both the training and validation datasets.



To ensure that no single feature overwhelmingly dictates predictions again, we re-ran the feature importance analysis. This time, it showed that 'people using at least basic sanitation services (% of population)' became the most influential predictor, contributing nearly 80% of the model’s decisions. Rather than removing this feature, which would discard useful information, we decided to transform it into categorical bins like low, medium, and high to retain its predictive power and also prevent the model from overly relying on one variable.

Hyperparameters Evaluated and Rationale

We tuned the following hyperparameters to study their impact:

Hyperparameter	Variation 1(Default RF)	Variation 2 (Limited Depth RF)	Variation 3 (Regularized RF)
n_estimators	50	50	50
max_depth	None(fully grown trees)	5	3
min_samples_split	2	2	100
min_samples_leaf	1	1	50

- Default RF: Used as a baseline with no restrictions on tree depth.
- Limited Depth RF: Introduced a max_depth of 5 to control tree size and improve generalization.
- Regularized RF: Further restricted tree depth (max_depth=3) and increased minimum samples per split and leaf nodes to force generalization.

The first variation used the default parameters, which allow trees to grow fully. It was selected to observe performance without constraints. The second variation introduced a maximum depth of 5, chosen to limit tree complexity, which helps to balance learning and generalization. The third variation applied stronger constraints, setting max_depth=3, and requiring at least 100 samples per split and 50 samples per leaf, it forces the model to generalize more effectively by reducing variance. This progression will help us to evaluate how different levels of regularization impact our predictive accuracy and generalization.

Model Performance Metrics and Justification

We chose our performance metrics to be RMSE and R^2 because they provide a clear assessment of model accuracy and generalization. RMSE will measure the magnitude of prediction errors in our target variable, making it easy to interpret how far predictions deviate from actual values. R^2 will evaluate how well the model explains variance in our target variable. The evaluate_model function trains a given random forest model and computes these metrics for both training and validation datasets. We can then identify potential overfitting or underfitting to help us select the best model variation.

```
RF Default - Train RMSE: 0.5019, Validation RMSE: 1.1743
RF Depth Limited - Train RMSE: 1.0297, Validation RMSE: 1.5541
RF Regularized - Train RMSE: 2.5473, Validation RMSE: 2.6775
              Model  Train RMSE  Validation RMSE  Train R2  Validation R2
0        RF Default    0.501918         1.174296  0.997108       0.979485
1  RF Depth Limited    1.029675         1.554058  0.987829       0.964070
2    RF Regularized    2.547297         2.677519  0.925509       0.893344
```

1. Default Random Forest (Overfitting Issue)

- The Default RF model achieved the lowest RMSE on the training set, which suggests that it learned the training data patterns exceptionally well. However, the gap between training R^2 (0.9971) and validation R^2 (0.9795) indicated overfitting.
- Overfitting occurs when the model memorizes the training data rather than learning generalizable patterns. This results in great training performance but significantly worse validation performance.
- The Validation RMSE (1.1743) was more than double the Training RMSE (0.5019), highlighting that the model did not generalize well to new data.
- This overfitting problem could be due to the fully grown trees, which allow the model to capture noise in the training data rather than useful predictive relationships.\

2. Limited Depth Random Forest (Partial Improvement in Generalization)

- The Depth Limited RF (max depth = 5) aimed to address overfitting by restricting tree depth, preventing the model from capturing too many specific details from the training data.
- The Training RMSE increased from 0.5019 (Default RF) to 1.0297, meaning the model was less sensitive to minor fluctuations in the training data.
- Validation RMSE increased to 1.5541, showing slightly reduced generalization error compared to Default RF, but it still had some overfitting as the training score remained significantly higher than validation score.
- This model performed better in real-world prediction scenarios but still struggled with overfitting, as the validation performance did not improve as much as expected.

3.Regularized Random Forest (Best Generalization Performance)

- The Regularized RF introduced stronger constraints by setting max depth = 3, requiring 100 samples per split and 50 samples per leaf.
- This forced the model to focus on only the most important patterns, reducing variance and increasing stability.

- The Training RMSE increased significantly to 2.5473, meaning the model no longer fit the training data too tightly.
- The Validation RMSE was 2.6775, which was much closer to the training RMSE compared to the other two models, showing that the model was learning general patterns rather than noise.
- The Validation R^2 (0.8933) was close to the Training R^2 (0.9255), indicating strong generalization. This means that even though the model was not as accurate in training, it was far more reliable in making predictions on new data.

Conclusion: Selecting the Best Model

- RF Default had an extremely low training RMSE and very high R^2, which indicates that it was overfitting by memorizing patterns in the training data instead of learning generalizable relationships, making it unsuitable for real-world predictions.
- RF Depth Limited introduced some constraints by limiting tree depth, which helped reduce overfitting and maintain strong predictive accuracy. However, it still retained a relatively high R^2 and a noticeable gap between training and validation RMSE, indicating that some overfitting persisted.
- RF Regularized, despite having the highest RMSE among the three variations, it actually demonstrated the most balanced performance across training and validation datasets. With stricter depth constraints and increasing the minimum number of samples per split and leaf, the model was forced to generalize better rather than memorize training data. This resulted in a lower training R^2 but a more reliable validation R^2, which suggests that the model could perform well in real-world settings.

Test Set Evaluation

To further verify that the Regularized RF model was the best choice, we evaluated it on the test dataset (2019-2021), which contained unseen data. Below is our results

Dataset	**RMSE**	**R^2**
Training	2.5473	0.9255
Validation	2.6952	0.8919
Test	2.3506	0.9173

The results showed that the test RMSE was lower than the validation RMSE, which indicates that the model could generalize well to new data. The test R^2 remained high, suggesting that the model maintained strong predictive power as it reduced reliance on patterns specific to the training and validation sets. The consistent trend across training , validation, and test datasets showed that the Regularized RF model was effective in balancing accuracy and generalization. This model provided a more stable and interpretable performance across all datasets, which makes it the most reliable choice for our forecasting.